PRACTICAL -8

1.Aim

Interface Fingerprint Sensor with Raspberry Pi.

2.Prior Concepts

i) Concept of Python Programming.

ii) SPI Communication interface.

3. New Concepts

Hardware connection of fingerprint sensor.

4. Hardware Required

i) USB to TTL converter.

ii)Fingerprint Module

iii) Jumper Wire

5.Procedure

Connect the finger print module to any one of the USB port of Raspi.

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Setting up Raspberry Pi for Serial Communication

sudo raspi-config

In the "Interfacing Options", select the "UART" option.

Select "Yes" option.

Finish the process and Reboot the Raspberry Pi.

Following are the terminal command that needs to be executed.

sudo apt update

pip install pyfingerprint

6. Implementation with Python Code

Code:

```
#l/usr/bin/env python_
#-- coding: utf-8--

import time
import RPi.GPIO as GPIO
import sys
from threading import Thread
from pyfingerprint.pyfingerprint import PyFingerprint

##############GP|O configuration#######
GPIO.setwarnings(False)
GPIO.cleanup()
GPIO.setmode (GPIO.BCM)
########################################

################### Global Variables ##############
#Initialize Fingerprint Module object
try:
        fingerprint_module = PyFingerprint('/dev/ttyUSBO', 57600,
OxFFFFFFFF,0x00000000)
        if (fingerprint_module.verifyPassword() == False ):
```

```
            raise ValueError('The given fingerprint sensor password is wrong!')
    except Exception as e:
            print('Exception message: '+ str(e))
            exit(1)

######################User function##############################
def delayms(time_in_msec):
    time.sleep( (time_in_msec/1000))
    pass

def delaysec(time_in_sec):
    time.sleep(time_in_sec)

def enrollFingerInDB():
    print ("Enroll your Finger into Fingerprint Database")

    ==========Start===========
try:
    print('Keep your finger...')

    # wait till the person keep finger on module
    while (fingerprint_module.readImage() == False ):
            pass
    # Transform read fingeprint image to characteristics and stores
    charbuffer 1
    fingerprint_module.convertlmage(0x01)

    # Checks for duplicate template in database
    result = fingerprint_module.searchTemplate()
    positionNumber = result[0]
```

```
        if (positionNumber >= 0):
                print('This fingerprint template already exists at position number = '
                +str(positionNumber))


        print('Remove your finger...')
        delaysec(2)


        print('Keep your finger again...')


        ## Wait that finger is read again
        while (fingerprint_module.readImage() == False ):
                pass


        ## Converts read image to characteristics and stores it in charbuffer 2
        fingerprint_module.convertImage(Ox02)


        ## Compares the charbuffers
        if (fingerprint_module.compareCharacteristics() == 0):
                raise Exception('Fingers do not match')


        ## Creates a template
        fingerprint_module.create Template()


        ## Saves template at new position number
        positionNumber = fingerprint_module.storeTemplate()
        print('Finger enrolled successfully!')
        print('New template position #'+ str(positionNumber))
except Exception as e:
        print('Enrollment fail.')
        print('Error:+ str(e))
         ==========End ==========
```

```
        pass


def searchFingerlnDB():
        print ("Search your Finger into Fingerprint Database")
#================Start==================
        try:
                print('Keep your finger...')
                ##Wait that finger is read
                while (fingerprint_module.readImage() == False ):
                        pass
                ##Converts read image to characteristics and stores it in charbuffer 1
                fingerprint_module.convertlmage(Ox01)


                ## Searchs template


                result = fingerprint_module.searchTemplate()


                positionNumber = result[0]
                 accuracyScore = result[1]


                if (positionNumber == -1):
                        print('No fingerprint template match found.')
                else:
                        print('Fingerprint found at position number ='+ str(positionNumbe


        except Exception as e:
                print('Searching fail.')
                print('Error:'+ str(e))
```

```
#==================End========
        pass

def deleteFingerinDB():
        print ("Delete your Finger data from Fingerprint Database")

#==============Start===========

try:
        positionNumber = input('Please enter the template position you want to delete (in range 0 to 1000): ')
        positionNumber = int(positionNumber)
        if (fingerprint_module.deleteTemplate(positionNumber) == True ):
                print('Template deleted successfully.')

except Exception as e:
        print('Deletion fail.')
        print('Error: '+ str(e))

#===========End===========
pass
def clearscreen():
        for i in range(5):
                print ("\n")
                pass
                pass

def clearscreen(no_of_line):
        for i in range(no_of_line):
                print ("\n")
                pass
```

```
pass


def getFingerprintTemplateCount():
        clearscreen(5)
        print('Fingerprint        template , in database = ‘
str(fingerprint_module.getTemplateCount())
str(fingerprint_module.getStorageCapacity()))       +’/
pass


## Tries to initialize the sensor


################## End of User function ##################
def main():
        clearscreen(5)
        choice = 0                  #for choosing the option
        global fingerprint_module #object for fingerprint module


print ("Initialization done...Starting program")
delaysec(1)


while True:                 #1st loop
        clearscreen(25)
        print( "\n\n Choose option in Menu below.\n 1. Enroll Fingerprint.\n Search
Fingerprint.\n 3. Delete Fingerprint.\n")
print (" 4. Get fingerprint record count in Database.\n 5. Exit Program.\n\n")
print (" Enter choice: ")
choice = int(input())


#Enroll finger into Database
if choice == 1:
        enrollFingerInDB()
```

```
        pass


#Search finger into Database.
        if choice == 2:
                searchFingerlnDB()
                pass

#Delete finger into Database.
        if choice == 3:
                deleteFingerlnDB()
                pass

#Get fingerprint templates database information
        if choice == 4:
                getFingerprintTemplateCount()
                pass

        if choice == 5:
                exit(0)
                pass
        delaysec(0.5)
        pass                            #end of 1st loop
pass
if name =='main ':
        try:
                main()
        except KeyboardInterrupt:
                pass
        finally:
                GPIO.cleanup()
```

```
print ("\n\nProgram terminated...")
```

7. Results